Physical AI

NVIDIA Cosmos

Develop physical AI faster with leading world foundation models and open data processing, training, and evaluation frameworks.

Download Models

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Cosmos 3

The Open Physical AI Foundation Model

The first omni-model with native reasoning, world and action generation. Built on Mixture-of-Transformers.

0:00 / 0:05

0:00 / 0:14

Power Vision AI Reasoning

Use as a vision language model (VLM) to reason over objects, interactions, and intent across complex real-world scenarios.

For real-time alerts and dense captioning across quality inspection, public safety, traffic monitoring, logistics, and autonomous driving.

Build Policy Models

Accelerate robot policy learning with NVIDIA Cosmos™ 3 as the backbone for World Action Models (WAMs).

Post-train the generalized world foundation model on specialized camera and embodiment data. The policy model adapts pre-learned actions to specific tasks, domains, and behaviors at scale.

0:00 / 0:07

0:00 / 0:05

Simulate Worlds

Scale Synthetic Video Data

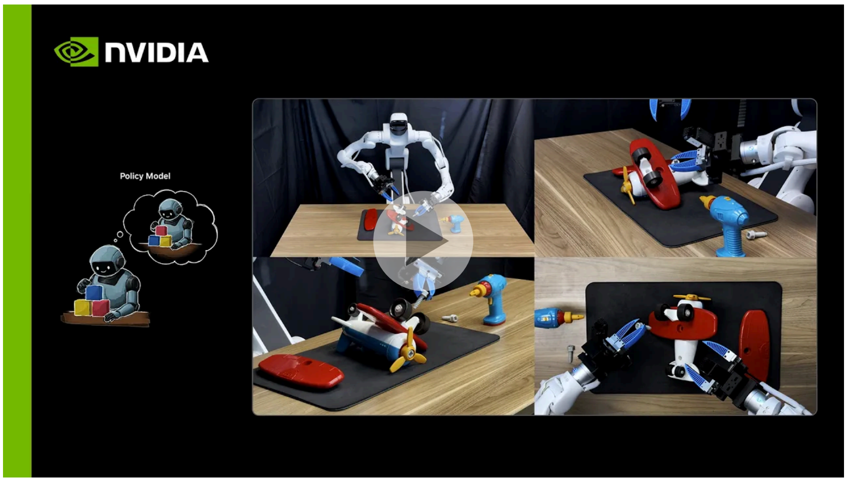
Scale the loop across environments, tasks, and conditions to continuously improve without real-world risk.

train physical AI without being constrained by what's been physically captured.

Learn More

Video

Introducing Cosmos



Meet NVIDIA Cosmos—the frontier of physical AI. Built on a new mixture-of-transformers architecture, this open omnimodel brings together pixels, actions, sound, and language to reason, plan, and generate what comes next.

Watch Video

Starting Options

Get Started With NVIDIA Cosmos

1

Ready to build? Access open models and code directly.

Download Models | Customize Models

2

Not ready to build yet? Try Cosmos models in our hosted catalog.

Need help? Start quickly with our hands-on model recipes.

Browse Cookbook

Develop With Cosmos

Build on the same technology powering Cosmos 3. Open frameworks and skills so developers worldwide can customize, extend, and contribute to physical AI.

Data Curation

Quickly filter, annotate, and deduplicate large amounts of sensor data with Cosmos Curator.

Download Cosmos Curator on GitHub

Review and score generative video outputs at scale using Cosmos Evaluator.

Download Cosmos Evaluator on GitHub

Training and Acceleration

Quickly build, post-train or deploy world models using open post-training, evaluation, optimization frameworks, and inference scripts and skills.

Start Building With GitHub

Agent Skills for Synthetic Data Generation

Turn coding agents into synthetic data experts for physical AI development.

Try Now

Use Cases

How Cosmos Accelerates AI Across Industries

Robot Learning Autonomous Vehicle Training Video Analytics AI Agents

Robot Learning

Build a robot learning policy that enables embodied agents to operate in real-world environments under both seen and unseen conditions.

See Examples

Performance

Runs Best On NVIDIA AI

Cosmos 3 is optimized for the best performance on NVIDIA hardware. NVIDIA RTX PRO™ 6000 Blackwell Series Servers accelerate physical AI development for robots, autonomous vehicles, and AI agents across training, synthetic data generation, simulation, and inference.

Unlock peak performance for Cosmos world foundation models on NVIDIA Blackwell GB200 for industrial post-training and inference workloads.

Learn More

Ecosystem

Adopted by Leading Physical AI Innovators

Next Steps

Join the Cosmos Community

Connect with Cosmos experts, engage with fellow developers, provide model feedback, and access continued learning through livestreams and recipes.

Join Now

Cosmos Cookbook

A comprehensive guide for working with the NVIDIA Cosmos ecosystem for real-world, domain-specific applications across robotics, simulation, autonomous systems, and physical scene understanding.

Learn More

NVIDIA Cosmos Lab

NVIDIA Cosmos Lab pioneers large-scale world foundation models that simulate the physical world — the foundation for breakthrough Physical AI.

Explore Now

Resources

The Latest From Cosmos Developers

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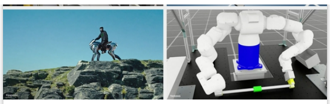
July 22, 2026

Make Long-Running NVIDIA TensorRT Engine Builds Observable and Cancelable in Python or C++

A TensorRT engine build can take seconds to many minutes. Large strongly typed models, deep tacti...



July 16, 2026

Japan's Robotics and Manufacturing Leaders Build on NVIDIA Cosmos to Advance Physical AI Frontier

NVIDIA today announced that Japan's physical AI leaders are building on the NVIDIA Cosmos™, NVIDIA...

Frequently Asked Questions

What's new in NVIDIA Cosmos architecture?

What is the licensing model for Cosmos world foundation models?

How do I post-train Cosmos models for my downstream applications?

Can I build a world model from scratch using tools from the Cosmos platform and my custom or in-house foundation model?

What's the difference between Cosmos 3 and previous Cosmos models?

What is the difference between Cosmos and Omniverse?